

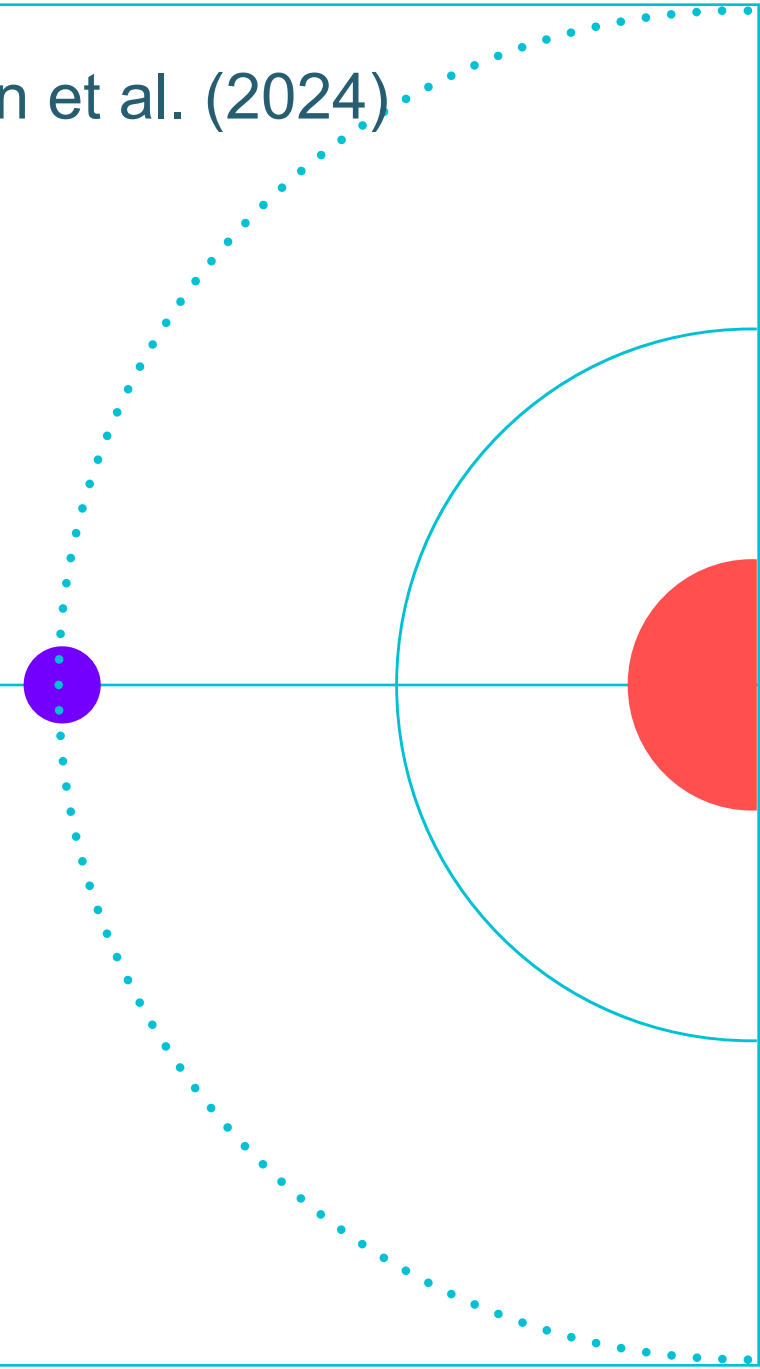
A Review of Software Logging Practices and Challenges

Based on Batoun et al. (2024).

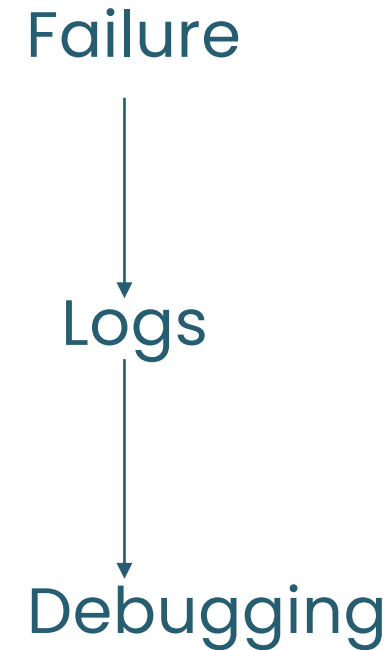
TUHH
Technische
Universität
Hamburg

Fallak Amin, 598871
Computer Science

Debugging -
Methodology
instead of
Confusion



- Hard-to-understand failures
- Logs capture runtime context
- Good logs support debugging and diagnosis
- Useful logging depends on where, what, and how much to log



Can you find the bug?

```
String env = Files.readString(Path.of("env.txt"));

Map<String, String> configs = Map.of(
    "dev", "dev-db.local",
    "prod", "prod-db.company.com"
);

if (!configs.containsKey(env)) {
    throw new RuntimeException("Database config not found");
}
```

Relevant Java code

```
Exception in thread "main"
java.lang.RuntimeException: Database config not found
```

Output

prod

env.txt

Why does this code fail even though prod exists?

Adding logs is not enough

```
System.out.println("Loading database config...");

if (!configs.containsKey(env)) {
    System.out.println("Database config missing.");
    throw new RuntimeException("Database config not found");
}
```

Relevant Java code

```
Loading database config...
Database config missing.
Exception in thread "main"
java.lang.RuntimeException: Database config not found
```

Output

Good logs show context

```
System.out.println("Requested environment: '" + env + "'");  
System.out.println("Environment length: " + env.length());  
System.out.println("Available configs: " + configs.keySet());
```

Relevant Java code

```
Requested environment: 'prod '  
Environment length: 5  
Available configs: [dev, prod]  
{...}
```

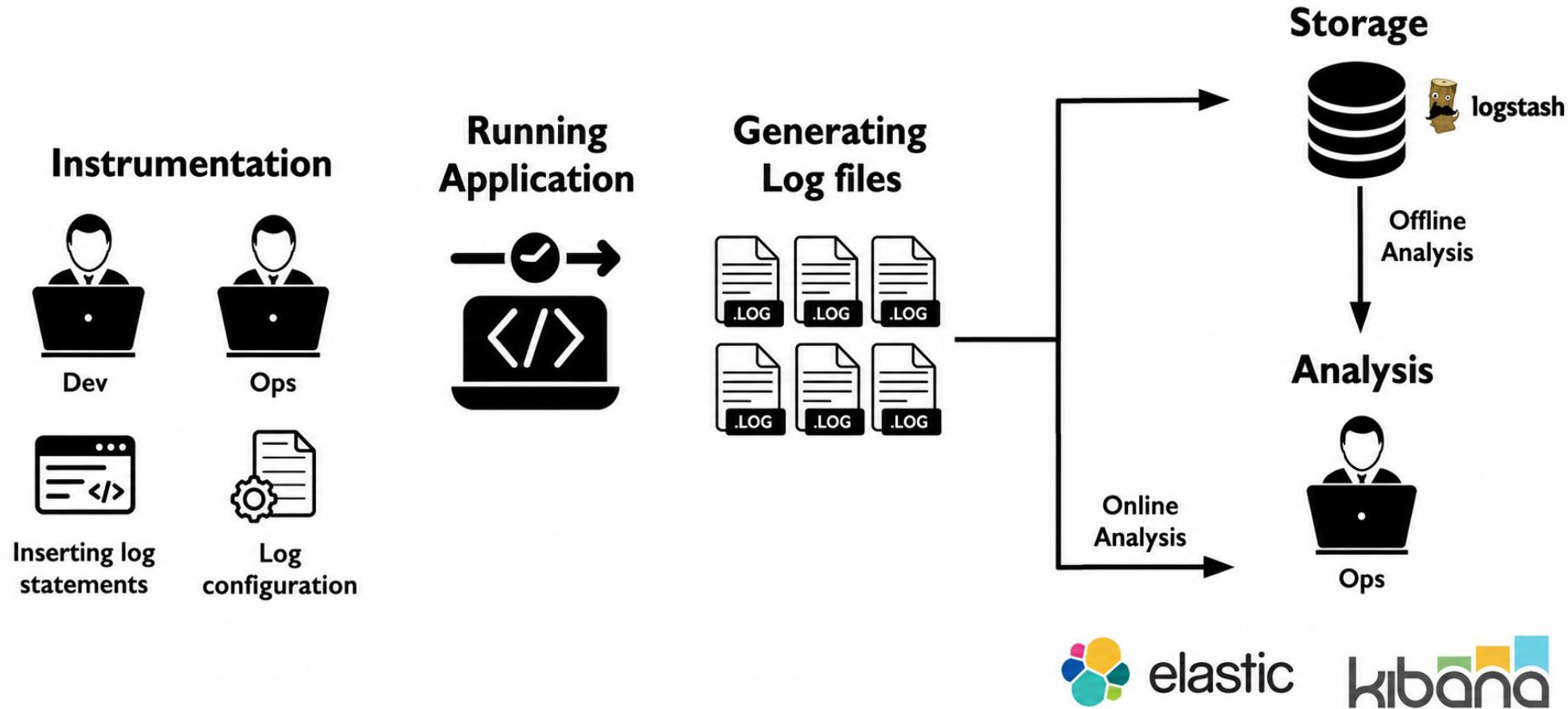
Output

- Logs reveal trailing whitespace in the text file
- -> use strip() to fix it

- Logs are useful only when they capture the right context
- Logging requires design decisions
- The paper studies logging from creation to analysis
- Systematic Literature Review of 204 papers

Logging Pipeline

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Source: Batoun et al. (2024), Fig. 1.

- Preparing software to produce logs
- Log statements in code
- Research focuses on location, context, verbosity level and configuration

- Categorizes logs based on purpose
- Log configuration controls logging behavior without changing code
 - Where logs are sent
 - How entries are formatted
 - Which levels are recorded

Debug

Info

Warning

Error

Critical

Setting the Log Verbosity Level

```
logger.debug("Raw env value: '{}', length={}", env, env.length());  
logger.info("Environment loaded");  
logger.error("No config found: '{}'", env);
```

```
<root level="DEBUG">
```

```
DEBUG Levels - Raw env value: 'prod ', length=5  
INFO  Levels - Environment loaded  
ERROR Levels - No config found for requested environment: 'prod '
```

```
<root level="INFO">
```

```
INFO  Levels - Environment loaded  
ERROR Levels - No config found for requested environment: 'prod '
```

- Storing logs in file or database
- Large logs with sensitive information -> storage must provide confidentiality, integrity, availability
- Different methods used, e.g. log pseudonymization and immutable log storage

- Replaces sensitive information with pseudonyms
- Protects confidentiality

2026-05-14 10:15:03 INFO User
logged in: alice@example.com
from IP 192.0.2.45



2026-05-14 10:15:03 INFO User
logged in: user_7F3A from IP
ip_91B2

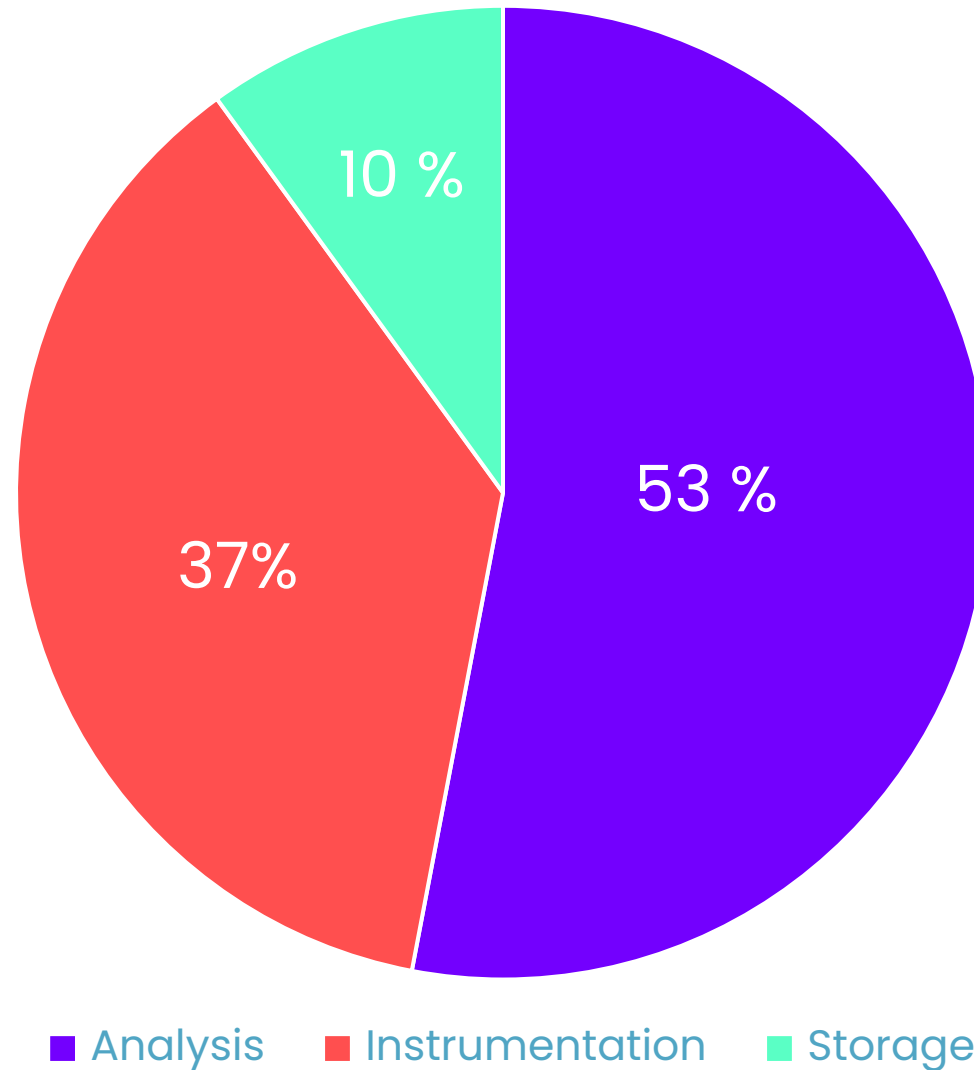
- Understand system behavior, debug failures, and diagnose problems
- Techniques include log parsing, template identification, clustering, and mining

- Converts raw logs into structured templates
- Separates fixed text from variables
- Enables mining, clustering, and automation

User 42 failed login from 10.0.0.5
User 77 failed login from 10.0.0.8



User <*> failed login from <*>



- Analyzed developer practice through StackOverflow
 - 20,766 logging questions quantitatively
 - 149 qualitatively
- 7 high-level and 22 fine-grained topics found
- 18,623 questions about log configuration

- 5 out of 7 StackOverflow high-level topics were not covered by the literature
- Research focuses mostly on improving log analysis
- Developers struggle with configurations and real-world usage

- StackOverflow does not represent all developer practice
- Beginner-focused, not industrial logging practices
- Research gap may be biased by data source

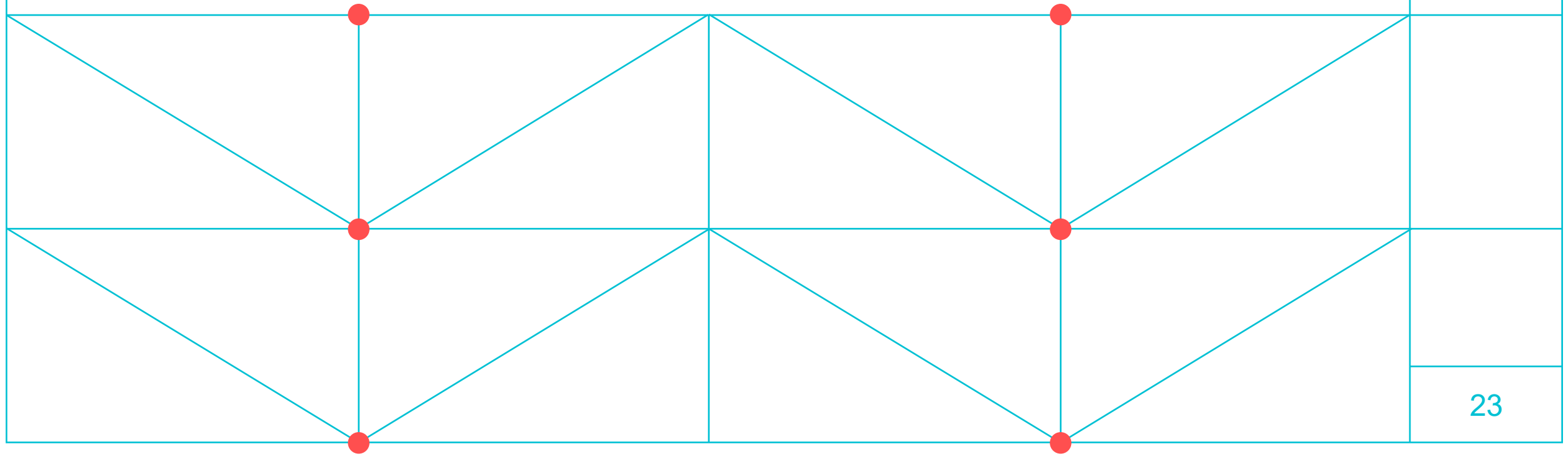
- Chen & Jiang (2021) and Rong et al. (2017) focus more on log instrumentation
- Mendes & Petrillo (2021) research log verbosity levels specifically
- El-Masri et al. (2020) clean logs for analysis through parsing and abstraction

- Logging is useful, but difficult to do well
- Research focuses mainly on analysis techniques
- Developers struggle mainly with configuration and practical usage
- Future research should address real-world logging problems more directly

- Batoun, M. A., Sayagh, M., Aghili, R., Ouni, A., & Li, H. (2024). *A literature review and existing challenges on software logging practices: From the creation to the analysis of software logs*. Empirical Software Engineering, 29, 103. <https://doi.org/10.1007/s10664-024-10452-w>
- Chen, B., & Jiang, Z. M. (2021). *A survey of software log instrumentation*. ACM Computing Surveys, 54(4), 1–34. <https://doi.org/10.1145/3448976>
- Rong, G., Zhang, Q., Liu, X., & Gu, S. (2017). *A systematic review of logging practice in software engineering*. In 2017 24th Asia-Pacific Software Engineering Conference (APSEC), 534–539. <https://doi.org/10.1109/APSEC.2017.61>
- Mendes, E., & Petrillo, F. (2021). *Log severity levels matter: A multivocal mapping*. In 2021 IEEE 21st International Conference on Software Quality, Reliability and Security (QRS), 1002–1013. <https://doi.org/10.1109/QRS54544.2021.00109>
- El-Masri, D., Petrillo, F., Guéhéneuc, Y.-G., Hamou-Lhadj, A., & Bouziane, A. (2020). *A systematic literature review on automated log abstraction techniques*. Information and Software Technology, 122, 106276. <https://doi.org/10.1016/j.infsof.2020.106276>

- What are problems with logging that you are struggling with yourself?

Bonus Slides



- 35 % of log levels were inconsistent with the developers' intention
- Automated tools that help with where, what, and at which level to log
- Analysis tools to detect bad logs

- Log compression: up to 94.98% compression rate
- Secure log storage: blockchain-based immutable log storage etc.
- Developers asked about rotating logs and log files overwriting each other

- Log template identification: ML-based, vectorization, clustering, etc.
- Mining: improve performance and speed

- Handling log files
- Dependency configuration of logging libraries
- Infrastructure-related configuration
- **Usage of logging libraries in the code**
- Scattered logging
- **Parsing complex logging data**
- Context-dependent usage of logs